



Bowling ARena



Goal : Create a AR Bowling

- Calibration of 2 cameras at the same time in Python.
- Ball detection in both cameras in Python.
- Triangulation of the ball's position.
- Forward results in Unity.
- Get the scene with a last camera for Unity.
- Evolution of the ball in the virtual world in Real-time.
- Interaction between the ball and the virtual objects.

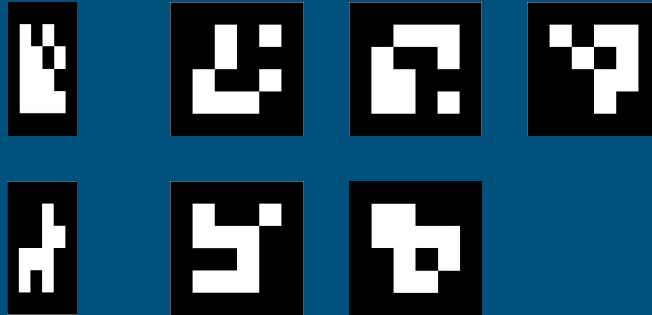
Mise en place

- Placement of Cameras, ArUco tags and origin of World
- Get the positions of each corner of the ArUco tags in the real world
- Each camera need to get every ArUco tags

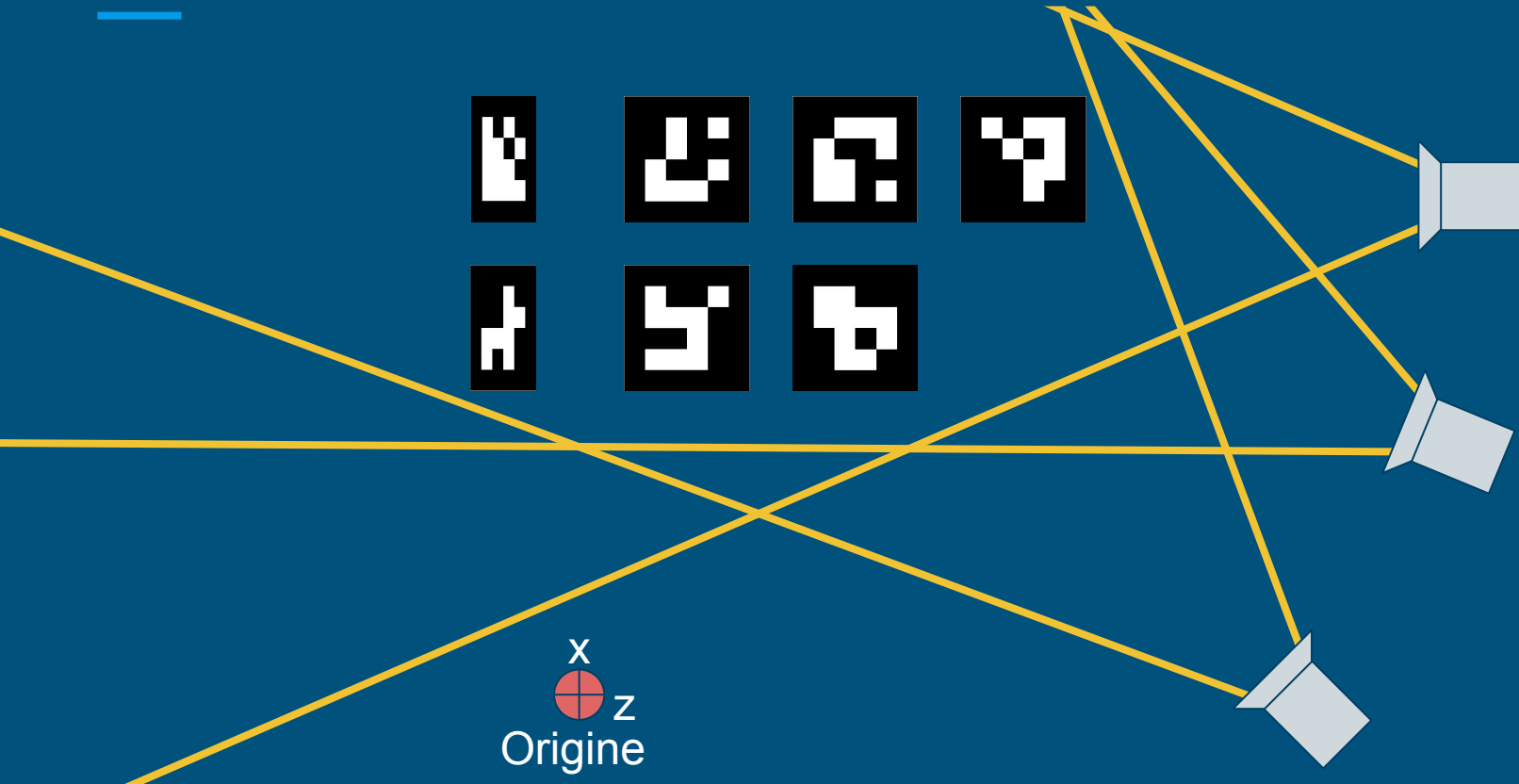
Mise en place - Cameras & ArUco codes



Mise en place - World origin



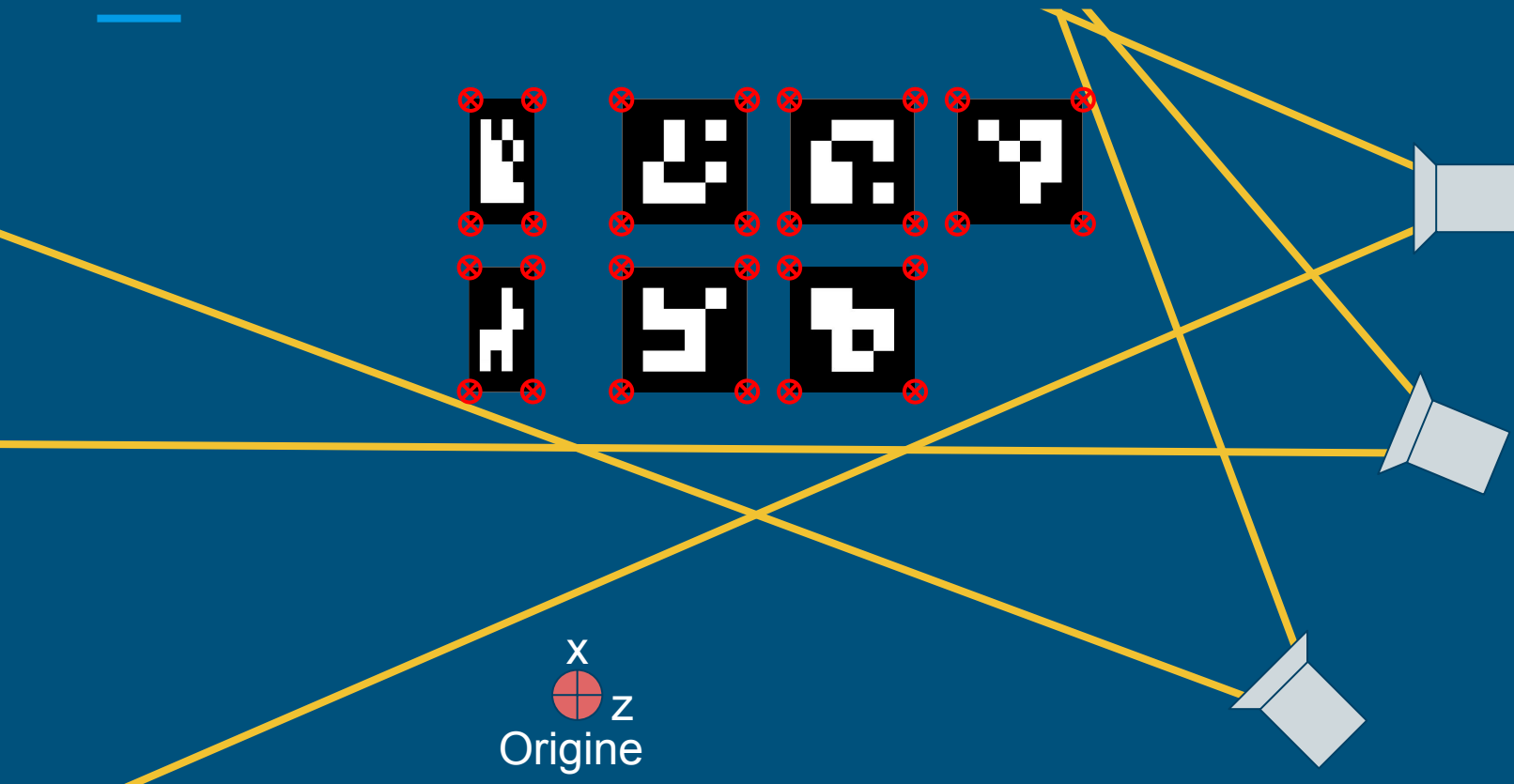
Mise en place - Cameras placement



Calibration

- Detection of ArUco tags with OpenCV
- Solve ArUco tags order
- Calibration of each cameras with OpenCV *solvePnP()*
- Cameras only need one calibration

Calibration - Detection & Calibration



Calibration - Detection & Calibration

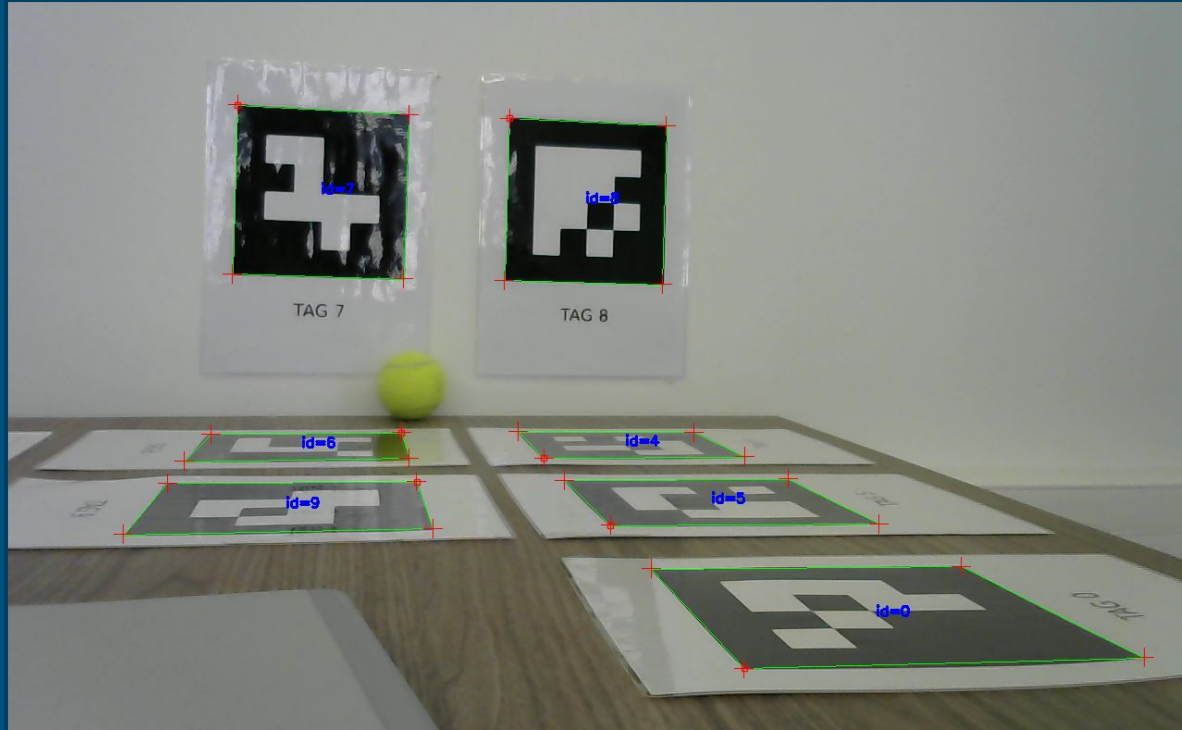


figure 1 - Calibrage d'une caméra

Ball detection

- Use of mask to extract the yellow.
- Image rendered in grayscale. -> Image 3
- Thresholding by intensity on the image to remove unintended artifacts.
- Circle detection with the Hough method.
- Determination of the circle with the largest radius.
- Getting the center of this circle. -> Image 4

Détection de la balle



Image 2 - Image originale



Image 4 - Détection du centre de la balle

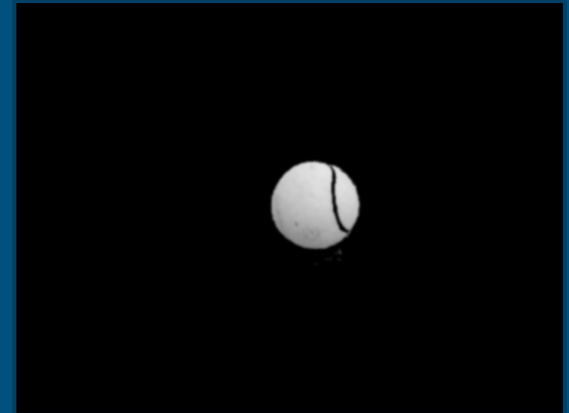
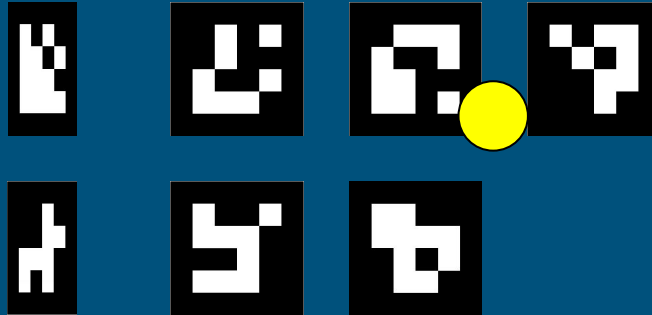


Image 3 - Image masqué en niveau de gris

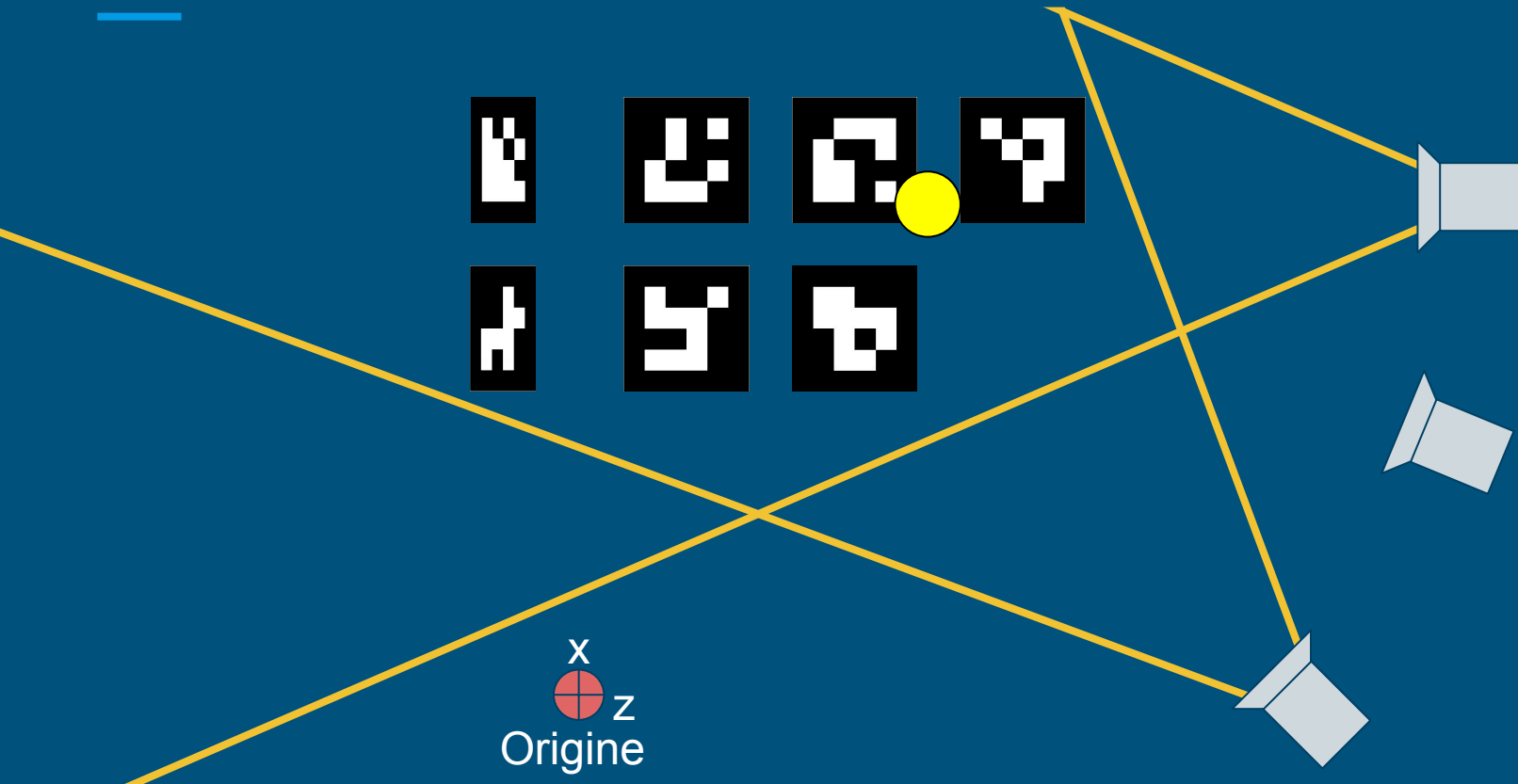
Triangulation of the ball's position

- Getting the projection matrices for each camera.
- Removing point distortion.
- Use OpenCV *triangulatePoints()* which uses the projection matrices and the ball's positions in pixels to triangulate the ball's position in the real world.
- Converting from homogeneous coordinates to Euclidean coordinates.
- Getting the ball's position in the real world.

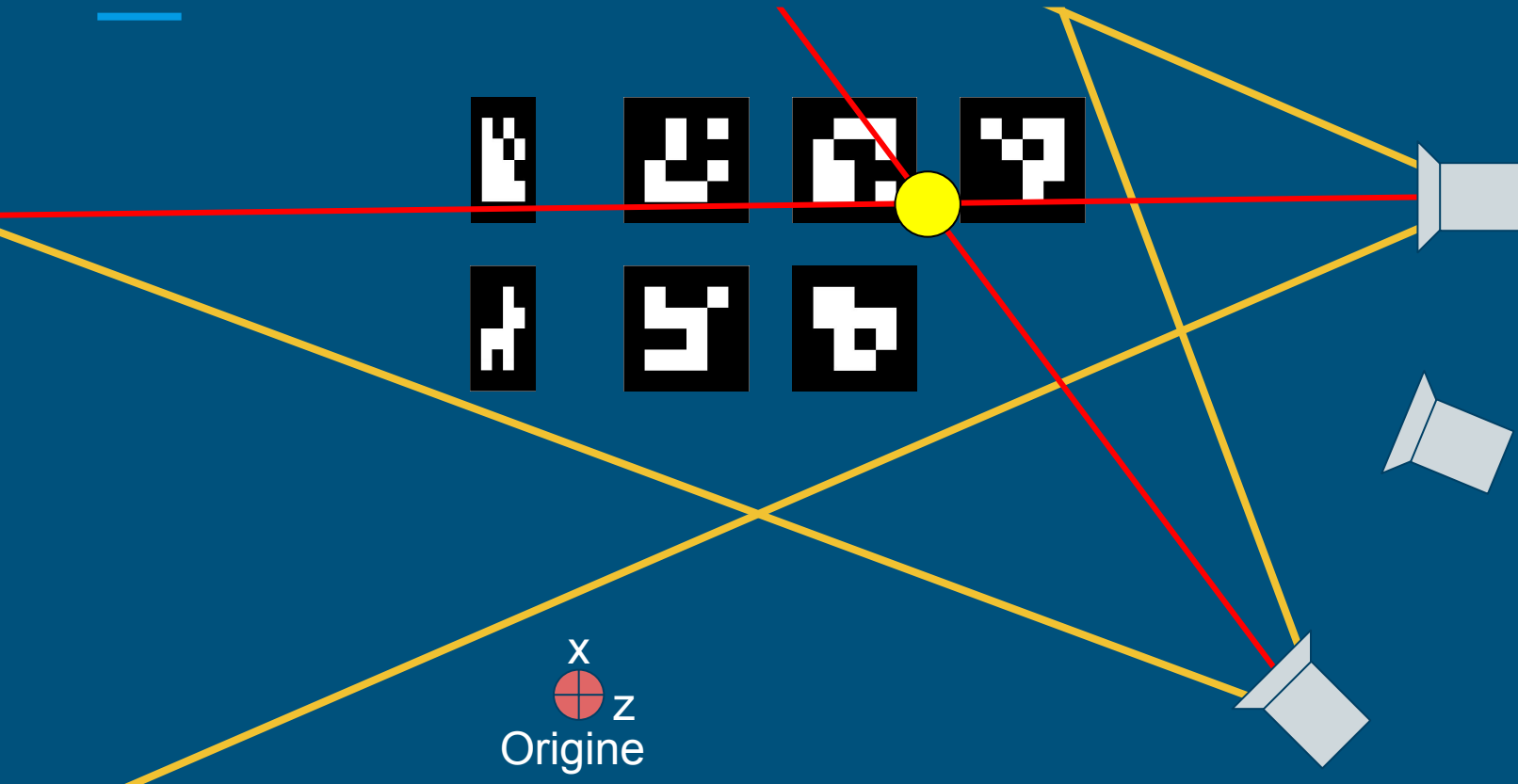
Triangulation



Triangulation



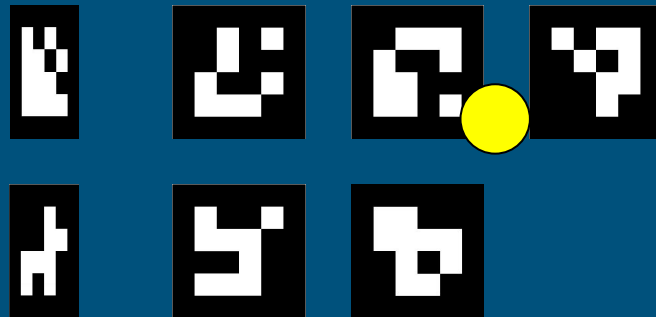
Triangulation



Forward result in Unity

- Usage of UDP function to link Python with Unity.
- Sending the position of the ball and the rotation and translation information on the camera.
- Get AR by Retrieving the video stream with a third camera to display in Unity.

Forward result in Unity



Unity



Real-time ball tracking

- Reproduction of the real world in the virtual world on a large scale
- Triangulation and real time ball's position.
- The ball in the real world is represented by a sphere in the Unity world with collisions enabled.
- Video render camera calibration.

Unity

- Handling collisions over an imported texture



cylinder influenced by physics

is parent of



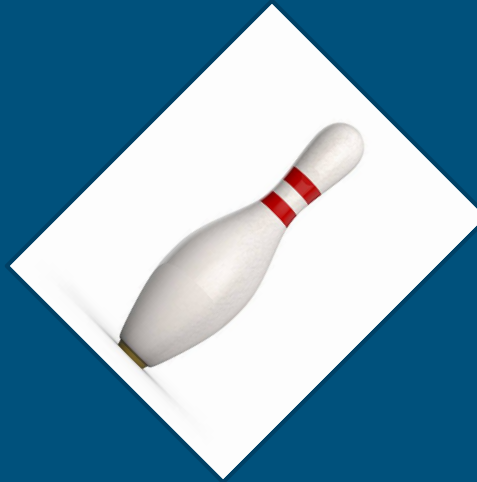
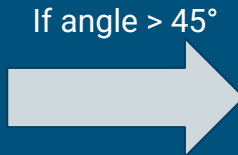
Pin texture without hitbox

Unity

- Handling collisions over an imported texture
- Victory condition management



Standing pin

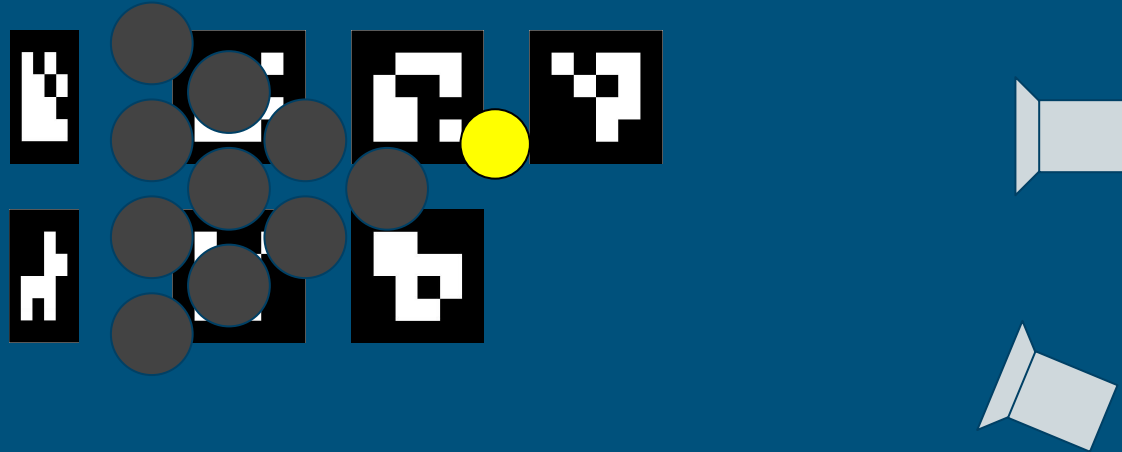


Fallen pin

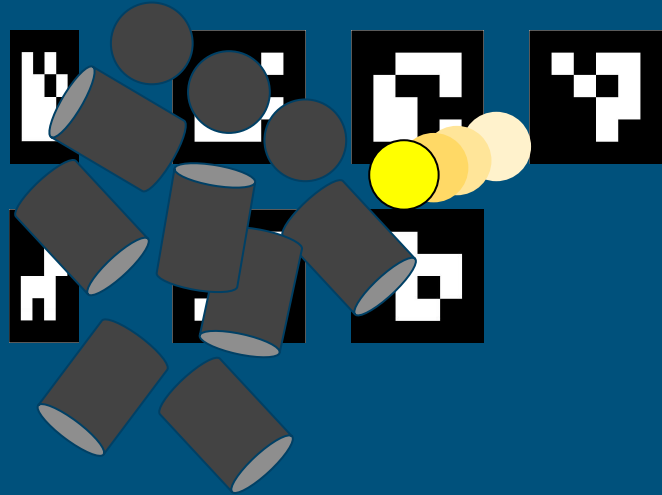
Unity

- Handling collisions over an imported texture
- Victory condition management
- Game restart management
- Managing text in the world
- Handling collisions between the pins and the ball

Final result



Final result



Final result

